

Geometry Unit 3 Assessment Guide

General Information	Throughout units 1 and 2, students have built a foundation of angles, parallel lines, and proofs. This foundation was often an informal exploration. Unit 3 balances the informal exploration of parallel lines with the more formal relationships and proofs. Lessons 8 and 9 broaden a student's experience with formal proofs. Students will continue to complete proofs throughout the course, so teachers should expect student understanding of proofs to grow throughout the course. Unit 3's assessment includes only one formal proof because of the expected growth in a student's ability to complete proofs throughout the year. Also, in units 1 and 2, students have been applying their prior knowledge of solving equations. "Problem types include mathematical and real-world context where students evaluate the value of a variable that will make two lines parallel; utilize two sets of parallel lines or more than two parallel lines; or write and solve equations to determine an unknown segment length or angle measure." from the Geometry B.E.S.T. Instructional Guide for Mathematics page 14. This assessment includes problem types where students are writing and/or solving equations. Questions 3-5 provide students with different sets of parallel lines. The cognitive demand varies for each set of lines. Question 4 presents a straightforward set of lines assessing a student's basic understanding of the relationships. Teachers can use student responses to questions 3-5 to determine if the complexity of the provided lines confuses students. The complexity of lines matches what students experienced in the unit. Teachers should expect their students to apply their knowledge of relationships in intersecting lines to lines intersecting with varying levels of complexity. If students struggle with a variety, then teachers are encouraged to support students by revisiting some of the questions in the lessons that were more complex.
Calculator Information	All questions are calculator neutral. Students may be given the use of a calculator and it will not have influence on their ability to show their knowledge of the content.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.912.GR.1.1
☐ Yes ☐ No If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.		Recommended Scoring (5 Total Points)	Students may provide an informal justification such as "alternate interior angles". It is left to the teacher's discretion as to the expected thoroughness from a student.
		Additional Information for question 1	Teachers should be aware that statewide assessments do not require students to formally write out the wording of postulates/theorems.
Question	1b	Benchmark(s)	MA.912.GR.1.1
☐ Yes ☐ No If two lines are intersected by a transversal and the consecutive interior angles are supplementary, then the lines are parallel.		Recommended Scoring (5 Total Points)	Students may provide an informal justification such as "same side interior angles". It is left to the teacher's discretion as to the expected thoroughness from a student.

Question	1c	Benchmark(s)	MA.912.GR.1.1
☐ Yes ☐ No The consecutive exterior angles must be supplementary angles to apply the converse of the consecutive exterior angles' theorem.		Recommended Scoring (5 Total Points)	Students may provide an informal justification such as "same side exterior". Since this is not a true statement, students would need to further explain that these angles do not fit the expected relationship of same-side exterior angles. It is left to the teacher's discretion as to the expected thoroughness from a student.
Question	1d	Benchmark(s)	MA.912.GR.1.1
☐ Yes ☐ No If two lines are intersected by a transversal and the corresponding angles are congruent, then the lines are parallel. If two angles are congruent by definition of congruence, they have the same measure.		Recommended Scoring (5 Total Points)	Students may provide an informal justification such as "corresponding". It is left to the teacher's discretion as to the expected thoroughness from a student.

Question	2	Benchmark(s)	MA.912.GR.1.1
	☐ $m\angle XTM = m\angle KRC$ ☐ $3x + 11 + 2x + 9 = 90^\circ$ ☐ $11x - 27 + 2x + 9 = 180^\circ$ ☐ $m\angle XTM + m\angle CYW = 180^\circ$ ☐ $11x - 27 + 3x + 11 + 2x + 9 = 180^\circ$	Recommended Scoring (15 Total Points)	Consider giving students 5 points for each correct choice. Students who include option 3 and 5 are not showing an understanding of congruent supplements and complements. Consider deducting 7 points if students have the three correct options and one incorrect option. Students who choose all 5 should receive no points.
Question	3a	Benchmark(s)	MA.912.GR.1.1
	$m\angle KYB = 39^\circ$	Recommended Scoring (5 Total Points)	Partial credit is not suggested.
Question	3b	Benchmark(s)	MA.912.GR.1.1
	$m\angle XLF = 41^\circ$	Recommended Scoring (5 Total Points)	Partial credit is not suggested.
Question	4a	Benchmark(s)	MA.912.GR.1.1
	$m\angle WPN = 108^\circ$	Recommended Scoring (5 Total Points)	Partial credit is not suggested.

Question	4b	Benchmark(s)	MA.912.GR.1.1
$m\angle LPW = 72^\circ$		Recommended Scoring (5 Total Points)	Partial credit is not suggested.
Question	5a	Benchmark(s)	MA.912.GR.1.1
$m\angle CWB = 138^\circ$		Recommended Scoring (5 Total Points)	Partial credit is not suggested.
Question	5b	Benchmark(s)	MA.912.GR.1.1
$m\angle HMY = 151^\circ$		Recommended Scoring (5 Total Points)	Partial credit is not suggested.
Question	6a	Benchmark(s)	MA.912.GR.1.1
15.6		Recommended Scoring (5 Total Points)	Students can find the value of x using the congruence of $\angle CNL$ and $\angle THR$ or students may set up a system of equations. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error.

Question	6b	Benchmark(s)	MA.912.GR.1.1
	10.5	Recommended Scoring (5 Total Points)	Students can find the value of y using a system of equations set up for question 6a or students can use that $\angle CNL$ or $\angle THR$ is supplementary to $\angle FBH$. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer. Consider deducting 2 points for such a minor error.

Question	7	Benchmark(s)	MA.912.GR.1.1												
<table><tr><th>Statement</th><th>Reason</th></tr><tr><td>1. $\overrightarrow{BH} \parallel \overrightarrow{RF}$</td><td>1. Given</td></tr><tr><td>2. $\angle HXW \cong \angle LYF$</td><td>2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.</td></tr><tr><td>3. $\angle HXW \cong \angle YWG$</td><td>3. Given</td></tr><tr><td>4. $\angle LYF \cong \angle YWG$</td><td>4. Transitive property</td></tr><tr><td>5. $\overrightarrow{RF} \parallel \overrightarrow{KG}$</td><td>5. If two lines and a transversal form corresponding angles that are congruent then the lines are parallel.</td></tr></table>		Statement	Reason	1. $\overrightarrow{BH} \parallel \overrightarrow{RF}$	1. Given	2. $\angle HXW \cong \angle LYF$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.	3. $\angle HXW \cong \angle YWG$	3. Given	4. $\angle LYF \cong \angle YWG$	4. Transitive property	5. $\overrightarrow{RF} \parallel \overrightarrow{KG}$	5. If two lines and a transversal form corresponding angles that are congruent then the lines are parallel.	Recommended Scoring (25 Total Points)	Consider distributing the points as 5 points for each missing statement or reason. Depending on the expectations of each teacher reason 4 could also be "Substitution". The teacher may also allow students to use shortcuts when writing the reasons for steps 2 and 5.
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		Additional Information	Teachers should be aware that statewide assessments do not require students to formally write out the wording of postulates/theorems.												